

Smart Factory – Vibration, OEE & RAG Short Guide

1. Vibration Monitoring (ISO 10816 / 20816 – Simplified)

Purpose: Classify machine vibration severity and trigger alarms.

Velocity RMS (mm/s) – Typical Zones:

Zone A (Good): < 2.8 mm/s – Normal operation

Zone B (Acceptable): 2.8 – 4.5 mm/s – Monitor trend

Zone C (Unsatisfactory): 4.5 – 7.1 mm/s – Alarm, plan maintenance

Zone D (Unacceptable): > 7.1 mm/s – Immediate action required

Used for motors, pumps, fans, compressors.

2. Common Vibration Root Causes

- Bearing wear: gradual RMS increase, high-frequency noise
- Misalignment: axial vibration increase
- Imbalance: dominant 1x RPM peak
- Looseness: broadband vibration increase

3. OEE (Overall Equipment Effectiveness)

OEE = Availability × Performance × Quality

Availability = (Planned Time – Downtime) / Planned Time

Performance = Actual Output / Ideal Output

Quality = Good Units / Total Units

High vibration → unplanned downtime → Availability loss.

4. RAG Usage Example

Input: Vibration RMS = 6.2 mm/s, Machine = Motor

Retrieved knowledge: ISO vibration severity zones

LLM Output: 'Value falls in Zone C. Risk of bearing wear. OEE availability likely to decrease due to planned maintenance.'

5. Why This Works Well for RAG

- Clear numeric thresholds
- Cause–effect explanations
- Business impact (OEE) linkage
- No hallucination risk when grounded in rules